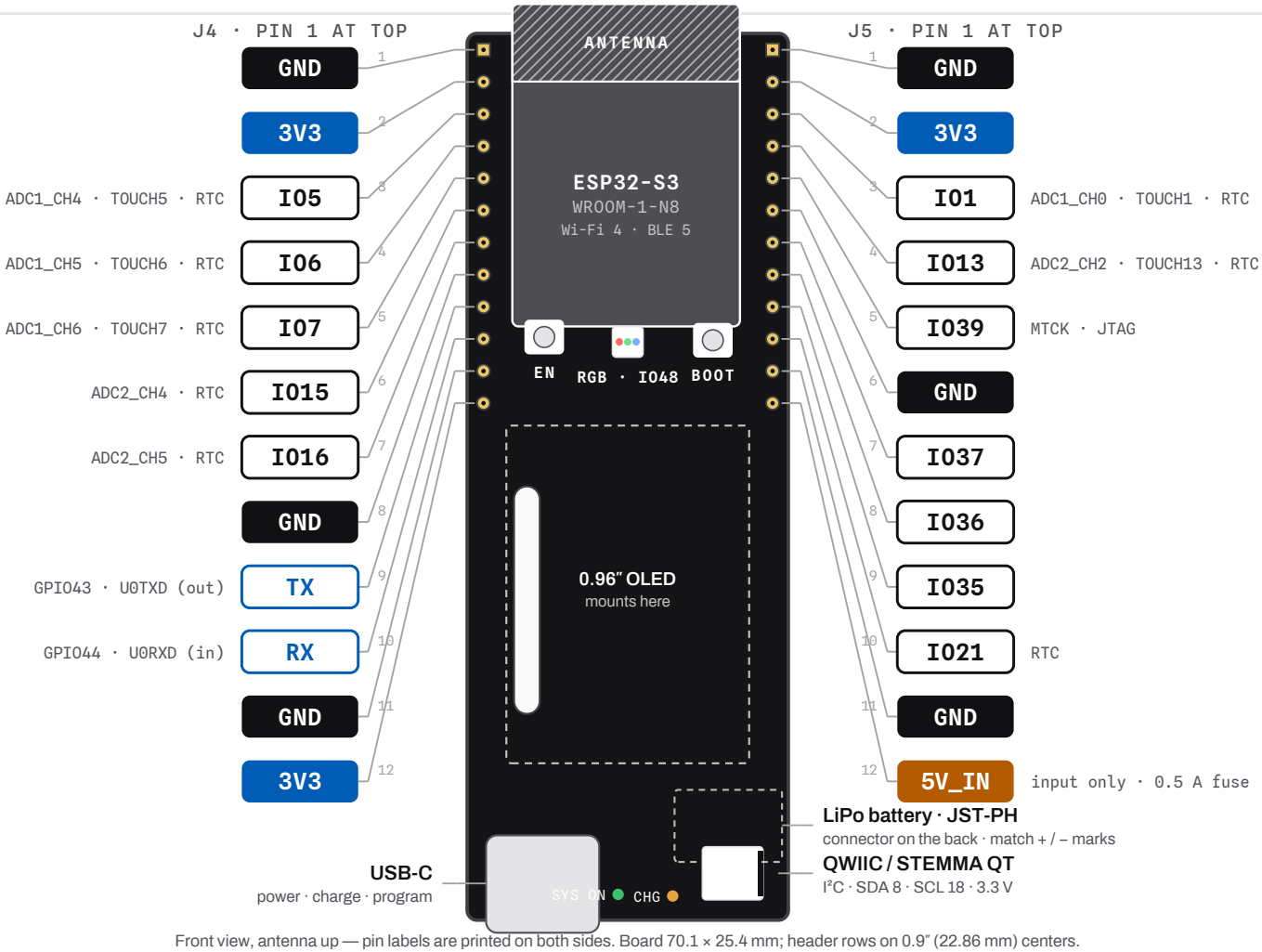
# ESP32-S3 DEV BOARD

PINOUT REFERENCE

**HW DESIGNS**
docs.hwdesigns.us




Front view, antenna up — pin labels are printed on both sides. Board 70.1 × 25.4 mm; header rows on 0.9" (22.86 mm) centers.

## ON-BOARD CONNECTIONS

| | |
|---|---|
| **OLED · SPI** | CS 10 · DC 9 · RST 14 · SCK 12 · MOSI 11 |
| **Qwiic · I²C** | SDA 8 · SCL 18 · 4.7k pull-ups |
| **RGB LED** | IO48 (SK6812) |
| **BOOT button** | IO0 — user key after boot |
| **USB D- / D+** | IO19 / IO20 |
| **VBAT sense** | IO4 = battery voltage / 2 |
| **VBUS sense** | IO2 = USB voltage / 2 |

These signals are not brought out to the headers.

## GOOD TO KNOW

All I/O is 3.3 V logic — pins are not 5 V tolerant.
Keep each pin under ~20 mA.
ADC2 (IO13, IO15, IO16) is unavailable while Wi-Fi is active — use ADC1 pins (IO1, IO5–IO7) instead.
IO35–IO37 are fully free on this module (N8, no PSRAM).
IO39 is JTAG MTCK by default; still works as GPIO.
RTC pins can wake the chip from deep sleep.
Any pin can be UART / I²C / SPI / PWM (GPIO matrix).

## POWER

Inputs: USB-C · 1-cell LiPo (JST-PH) · 5V_IN pin.
5V_IN is input only (0.5 A fused) — it never outputs 5 V. Unplug the battery when powering from 5V_IN.
3V3 pins are regulator outputs — budget ≈ 150 mA for your add-ons.
Battery charges from USB-C only, at ≈ 300 mA.
CHG LED: on = charging · off = full or no battery.

GND · 3V3 · 5V_IN · IOxx · UART


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